

SIMATS ENGINEERING

Name	Register Number	Course Code	Course
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40. Draw Rectangular shape and extract objects.

AIM:

Draw Rectangular shape and extract objects.

PROGRAM:

```
import cv2

# Read the image
img = cv2.imread(r"C:\Users\Susmitha\Downloads\ITA0510 Lab\exp1.png")

if img is None:
    print("Error: Image not found!")
else:
    # Define rectangle coordinates
    start_point = (50, 50)
    end_point = (200, 200)

    # Rectangle color (Red in BGR)
    color = (0, 0, 255)

    # Thickness of rectangle
    thickness = 2

    # Draw rectangle
    rect_img = cv2.rectangle(img.copy(), start_point, end_point, color, thickness)

    # Display image with rectangle
    cv2.imshow("Image with Rectangle", rect_img)

    # Extract the object inside the rectangle
    obj_img = img[start_point[1]:end_point[1], start_point[0]:end_point[0]]

    # Display extracted object
    cv2.imshow("Extracted Object", obj_img)

    # Save extracted object
```

```
cv2.imwrite(r"C:\Users\Susmitha\Downloads\ITA0510 Lab\object.jpg", obj_img)
```

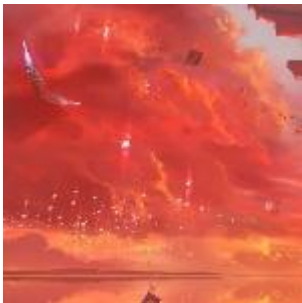
```
cv2.waitKey(0)
```

```
cv2.destroyAllWindows()
```

INPUT:



OUTPUT:



RESULT:

Thus the program is executed successfully.